

GlucoMon

Non-Invasive Optical Blood Glucose Monitoring Wearable Device

Semifinal Report
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Team AC/DC

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1. Introduction and Project Overview

Diabetes is one of the fastest-growing non-communicable diseases worldwide, and keeping blood glucose within a healthy range is central to preventing complications such as cardiovascular disease, kidney failure, nerve damage and vision loss. Continuous glucose monitoring has therefore become a key part of modern diabetes care.

Conventional monitoring methods, however, come with real drawbacks. Finger-prick glucometers are accurate but invasive and painful with repeated use, and implantable Continuous Glucose Monitoring (CGM) systems are expensive and require periodic sensor replacement. These limitations reduce long-term adherence, particularly for patients who need to test several times a day.

GlucoMonitor is our response to this gap: a non-invasive blood glucose monitor based on green/IR-red light photoplethysmography (PPG) combined with a cloud-based machine learning model. The device estimates glucose levels from optical blood-flow signals captured at the fingertip, without any blood sampling, and displays the result on a small OLED screen.

2. Literature Review and Background Research

Our approach is grounded in a growing body of research showing that PPG signals carry measurable, glucose-correlated information. A key reference study collected PPG signals from 80 participants under fasting and post-glucose conditions, producing 160 paired PPG-glucometer profiles, and evaluated several regression techniques: Random Forest Regression, Support Vector Machines, K-Nearest Neighbours and Decision Trees, for estimating glucose non-invasively from these signals.

Further literature on Cloud-based PPG glucose estimation confirms that deep learning models can be compressed and deployed on microcontroller-class hardware while retaining useful accuracy, which directly supports the feasibility of running inference on an ESP32.

For model selection, we are evaluating existing open, state-of-the-art models from the following sources rather than training a model from scratch for this round:

- Zeynali et al., “Non-invasive blood glucose monitoring using PPG signals with various deep learning models and implementation using TinyML,” Scientific Reports (2025).
- Khan et al., “Non-invasive blood glucose estimation using green light photoplethysmography and machine learning,” Frontiers in Digital Health (2026).
- Open-source reference implementation:
“Predict-Blood-Glucose-Level-Based-on-PPG-Signals” (GitHub, yasirsaleem502).

Broader literature on CGM sensor principles and core technologies was also reviewed to understand the performance benchmarks (accuracy, lag, calibration needs) that a non-invasive alternative would approach.

2.1 Reported Performance of Candidate Models

Study / Model	Training & Test Data	Reported Performance	Tested Outside Training Data?
Khan et al. (2026) Random Forest Regression (best of 7 models tested)	80 subjects, 160 fasting/post-glucose PPG profiles; 50/50 split (80 train / 80 independent test)	$R^2 = 0.92$, MAE = 4.8 mg/dL, RMSE = 6.1 mg/dL	Yes, evaluated on an independent test set of subjects not seen during training
Zeynali et al. (2025) ResNet34 (1-second segmentation, best of 3 DL models)	VitalDB, 6,388 subjects (70% / 15% / 15% train-val-test split)	RMSE = 19.7 mg/dL, MAE = 14.8 mg/dL, Clarke Grid Zone A = 76.6%, Zone B = 23.4%	Yes, additionally tested on the separate MUST dataset (67 subjects, ages 0.3–94) to check generalisation to a new population
GitHub reference implementation (yasirsaleem502)	Smaller public PPG–glucose dataset	Repo-reported metrics not yet independently verified by our team	To be confirmed before adoption

Two points are worth highlighting. First, the Khan et al. RFR model is the more directly relevant reference for our hardware: it was built from a low-cost Arduino-based PPG sensor rather than clinical equipment, and it reports the strongest accuracy ($R^2 = 0.92$) of the three sources, though on a smaller 80-subject cohort. Second, the Zeynali et al. model trades some accuracy (RMSE 19.7 mg/dL vs 6.1 mg/dL) for a much larger and more diverse training set (6,388 subjects) and proven embedded deployment on a microcontroller, making it the stronger reference for the TinyML deployment side of our design, even though it was demonstrated on an STM32H743IIT6 rather than our ESP32 (a porting/feasibility check is needed.)

2.2 Validating Models Outside Their Training Data

Both peer-reviewed studies strengthens our feasibility claims: Khan et al. evaluate their RFR model on a held-out test set of subjects never seen during training, and Zeynali et al. go a step further by testing their best model on the entirely separate MUST dataset (67 subjects, ages 0.3–94) that was not used anywhere in training, specifically to check that the model generalises to a new population rather than just memorising the training cohort.

We plan to follow the same precedent with our own hardware. Importantly, both cited PPG datasets were collected with sensors different from ours — Khan et al. used a green-LED Pulse Sensor (SEN-11574), and the VitalDB signals used by Zeynali et al. came from clinical-grade equipment — whereas GlucoMonitor uses a dual-wavelength Red/IR MAX30102. PPG waveform shape, amplitude and noise characteristics vary across sensors, so a model trained on one device's signal is not guaranteed to transfer cleanly to another without recalibration.

train.csv		×	+
File	Edit	View	
1.0566,	123.1		
0.4348,	212.4		
0.6788,	178.5		
0.877,	157.8		
1.2112,	89.2		
1.232,	89.2		
1.3311,	74		
0.5868,	199.3		
0.7374,	158.2		
0.7764,	174.8		
1.4014,	73.5		
0.5147,	196.8		
0.6974,	180.7		
1.4428,	68.7		
1.2355,	102.5		
1.2427,	97.6		
1.119,	117.8		
0.7818,	169.3		

0.8358,	163.6
1.0475,	125.7
0.9238,	143.9
1.2116,	104.1
1.3197,	80.7
0.9068,	140.6
0.757,	172.9
1.0379,	121.3
1.1969,	96.6
0.8524,	149.2
1.135,	111.4
1.109,	110.1
1.0116,	123.4
1.3857,	69.5
0.4497,	208.5
0.5141,	200.7
0.6385,	188.7
1.1648,	103.1
1.3686,	76.5
0.4789,	203.4
0.5907,	185

Training Dataset CSV File (50 readings)

3. Feasibility

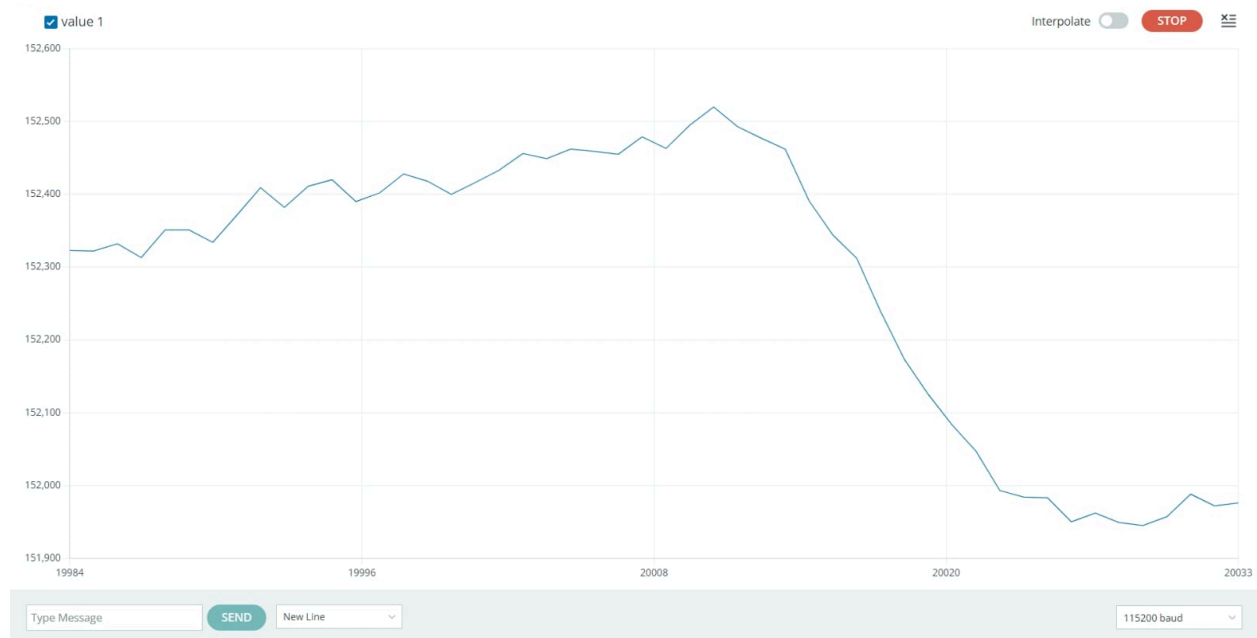
3.1 Component Availability

- MAX30102 dual-wavelength (Red/IR) pulse-oximetry and heart-rate sensor: widely available I²C sensor module used to capture red and infrared photoplethysmography (PPG) signals for non-invasive glucose estimation.
- ESP32-DevKitC-32D development board: dual-core MCU with built-in Wi-Fi/Bluetooth, used as the main controller.
- MPU6050 6-axis accelerometer/gyroscope: used for motion-artifact detection, shares the I²C bus.
- MicroSD card module — local logging of raw and processed signal data for offline analysis.
- 0.96" LCD display (128×64) — glucose readout, also on the shared I²C bus.
- LM2596 buck converter — steps down battery voltage to a stable supply rail for the sensor and logic.
- Rechargeable LiPo battery — portable power source sized for a wearable form factor.

The MAX30102, MPU6050, and 0.96-inch LCD display all communicate over the same I²C bus (SDA and SCL), minimizing the number of GPIO pins required on the ESP32-DevKitC-32D. The selected display screen will be upgraded in future developments. The MicroSD card module communicates over the SPI interface, providing high-speed data logging without interfering with the I²C devices.

3.2 Technical Feasibility

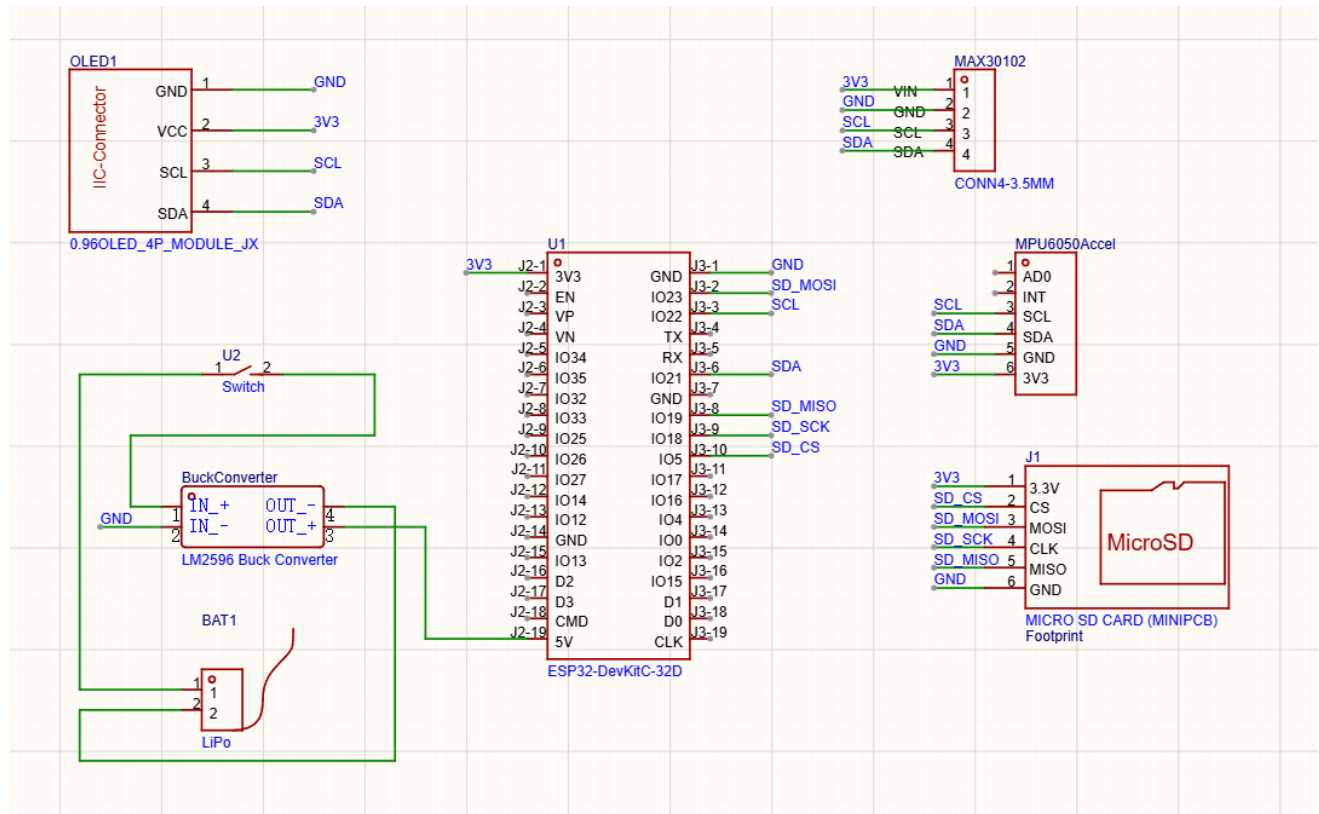
The technical approach rests on three pillars that the cited literature supports independently: (1) dual-wavelength PPG absorption differences correlate with blood-composition changes, including glucose-related variation; (2) classical regression and lightweight deep-learning models can map extracted PPG features to glucose estimates with reported accuracy in the clinically-relevant range; and (3) these models can be quantised to 8-bit integer form and run entirely on a microcontroller.



Sensor waveform of MAX30102

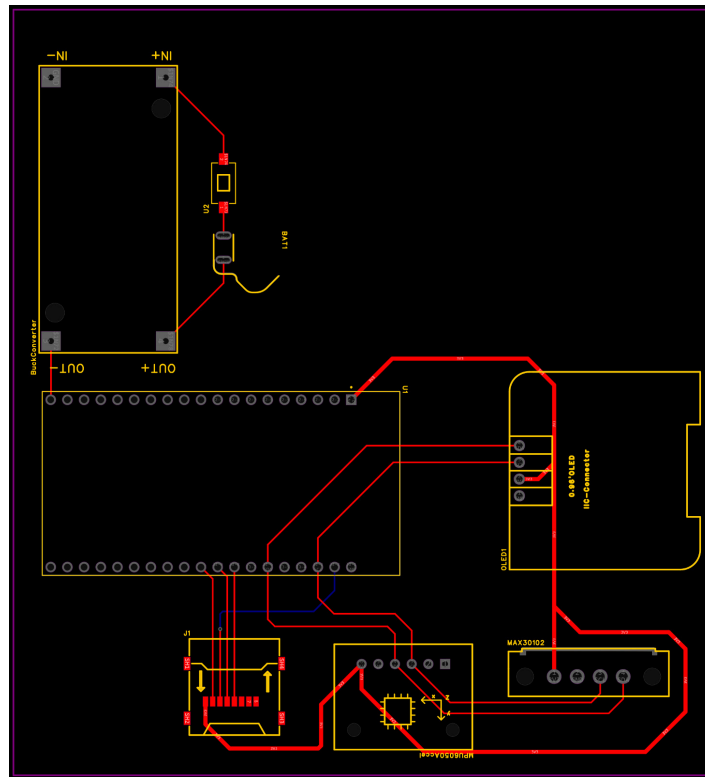
4. Technical Implementation

4.1 Circuit Design / Schematic



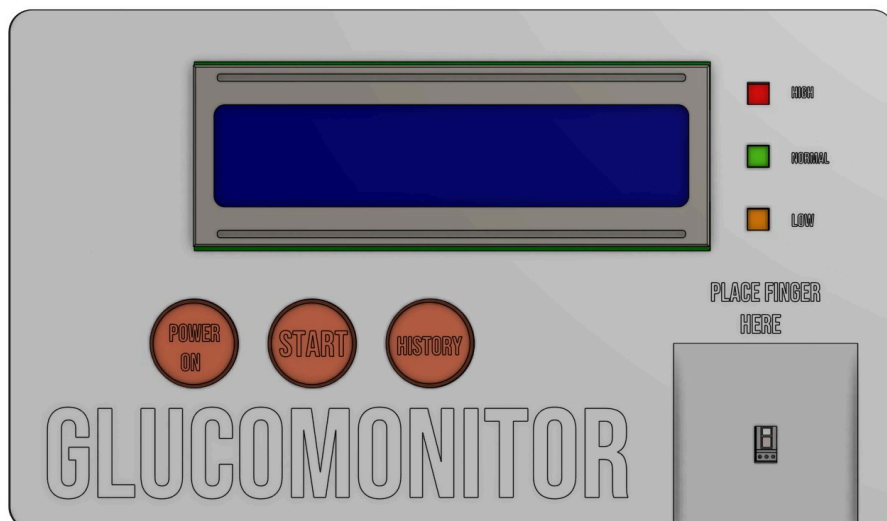
Glucomonitor circuit schematic (EasyEDA) — ESP32-DevKitC-32D with MAX30102 PPG sensor, MPU6050 IMU, MicroSD logging, LCD, and LM2596 buck-converter power supply.

4.2 PCB Design File



4.3 Enclosure Design

The housing is designed as a custom 3D-printed fingertip clip that holds the MAX30102 sensor flush against the skin. The enclosure uses opaque print material specifically to



achieve optical isolation — blocking ambient or solar light from reaching the photodiode, which is a common source of measurement error in wearable optical sensors.



Concept sketch of the fingertip-clip enclosure

The proposed device enclosure has been designed as a compact, custom 3D-printed desktop unit that integrates the sensing area, user interface, and internal electronics into a single housing. The top panel incorporates a display for presenting measurement results and system information, user control buttons for device operation, visual status indicators, and a dedicated finger placement area to promote consistent positioning during measurements.

The enclosure has been designed with a focus on usability, compactness, and ease of assembly, while providing sufficient internal space for the required electronic modules and wiring. The layout aims to facilitate straightforward operation while maintaining a clean and ergonomic appearance suitable for a portable point-of-care device.

At the current stage of the project, the enclosure design has been completed as a CAD model, representing the present progress of the prototype development. Fabrication and assembly of the physical prototype are currently underway.

GitHub Repository for our project: <https://github.com/amanda-ari/Glucomon.git>

5. Comparable Products Already in the Market

To support the claim that GlucoMonitor could realistically reach the market, we looked for two kinds of precedent: non-invasive glucose products already pursuing regulatory clearance, and products that use the same sensing and ML approach for a different measurement.

Product	Company	Sensing + ML Approach	Measurement	Regulatory Status
Hypertension Notification Feature	Apple	PPG + ML model trained on 100,000+ individuals	Hypertension risk	FDA-cleared, Sept. 2025
Hilo Band (formerly Aktiia G0)	Aktiia / Hilo	Cuffless optical PPG + AI	Continuous blood pressure	FDA 510(k) cleared, Jul. 2025 (OTC); CE since 2023; 130,000+ units sold
KnowU	Know Labs	Bio-RFID spectroscopy + LightGBM ML (same on-wearable ML pattern)	Non-invasive glucose	Not yet FDA-cleared; currently in clinical trials toward submission
D-Base / D-Pocket	DiaMonTech	Mid-infrared laser spectroscopy + ML	Non-invasive glucose	D-Base CE-certified; FDA submission in progress; D-Pocket targeted market-ready 2026/2027
Various fitness trackers / Galaxy-serie	Multiple OEMs	MAX30102-family PPG sensor	Heart rate, SpO ₂	Cleared as wellness or consumer features, not diagnostic

s phone sensors				
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The MAX30102-class PPG sensor is in the same sensor family already mass-produced for consumer heart-rate and SpO₂ features, so the hardware side of our design rides on already-proven, already-certified-for-adjacent-use components. Regulators have already cleared the general pattern our project depends on, a wearable PPG sensor feeding a machine-learning model to estimate a physiological value it cannot measure directly, for blood pressure (seen in Apple's Hypertension Notification Feature and the Hilo/Aktiia band), which is a meaningful precedent for an eventual glucose clearance pathway.

At the same time, it is important to be accurate about where the glucose-specific frontier actually stands: as of this writing, no non-invasive wearable (smartwatch, ring, or clip) has FDA marketing authorisation specifically for blood glucose, and the FDA has explicitly warned consumers against relying on any current non-invasive device for that purpose. Well-funded competitors such as Know Labs (NYSE-listed, 270+ patents) and DiaMonTech (over €25M raised, CE-certified D-Base) are still in clinical trials or pre-submission for glucose specifically. This confirms GlucoMonitor is aimed at a genuine, currently-open market gap rather than a solved problem which will set an honest expectation that clinical-grade non-invasive glucose accuracy is an industry-wide open challenge, not a shortfall unique to this prototype.

6. Data Security and Data Policy

- On-device inference means raw PPG waveforms never need to leave the device during normal operation — only the extracted features and the final glucose estimate are retained locally (MicroSD) or, in future, transmitted.
- The ESP32's built-in Wi-Fi/Bluetooth is reserved for over-the-air model updates and optional companion-app sync; any such channel will use encrypted transport (TLS/BLE pairing) rather than plaintext transmission.
- Locally logged data on the MicroSD card is intended for the user's own analysis/export and is not automatically uploaded anywhere.

7. Regulatory Aspects and Intellectual Property

As a healthcare-adjacent device, GlucoMonitor's real-world deployment will require regulatory clearance, including Sri Lankan and International regulations.

Aspect	Relevant Body	Status / Consideration
Device classification	FDA (US) / NMRA (Sri Lanka)	Glucose-monitoring devices are typically Class II medical devices, requiring demonstrated accuracy against a reference method before market authorisation.
Pre-market pathway (future)	FDA 510(k) / De Novo, NMRA registration	Outline of the validation studies, biocompatibility checks, and electrical safety testing.
Data protection	Sri Lanka Personal Data Protection Act / HIPAA	On-device inference limits transmission of raw physiological data; any future cloud sync must be encrypted and access-controlled.

8. Cost-Effectiveness

8.1 Target Market Price

Our pricing logic rests on the logic that patients already tolerate measurable accuracy trade-offs in exchange for comfort and lower cost, as the FreeStyle Libre vs. Dexcom market itself demonstrates.

Device	Type	Approx. Cost	Reported Accuracy (MARD)
Finger-prick glucometer + strips	Invasive, recurring	~LKR 50–100 per test (recurring)	~5–10%
FreeStyle Libre 3	Minimally-invasive CGM, recurring sensor	~USD 70/month (~LKR 23,500)	~9.2%
Dexcom G7	Minimally-invasive CGM, recurring sensor	~USD 350/month (~LKR 117,000)	~8.2%
GlucoMonitor (target)	Non-invasive, one-time purchase	~USD 45–65 one-time (~LKR 15,000–22,000)	Comparable PPG/ML studies report 12–20% MARD

8.2 Bill of Materials

All hardware components were chosen to be low-cost, commodity modules rather than specialised medical-grade parts, keeping the unit cost of a prototype well below comparable CGM systems.

Component	Qty	Est. Unit Cost (LKR)	Notes
MAX30102 PPG/Pulse-Ox sensor module	1	380.00	Core optical sensor
ESP32 NodeMCU-12S dev board	1	1400.00	Main controller
MPU6050 IMU/accelerometer module	1	680.00	Motion-artifact compensation
MicroSD card module + card	1	130.00	Local data logging

1602 16x2 Yellow Backlight LCD Display (DM0002)	1	400.00	Glucose value readout
LM2596 buck converter module	1	260.00	Power regulation
Li-ion battery + connector	1	300.00	Rechargeable power source
3D-printed enclosure (fingertip clip)	1	1500.00	Material + print cost
PCB fabrication (semifinal: optional)	1	1500.00	Not required for this round
Total (per unit, prototype)	-	6550.00	Excludes one-off tooling/labour

Unlike finger-prick glucometers (recurring test-strip and lancet costs) or CGM systems (recurring sensor replacement every 10–14 days), GlucoMonitor has no disposable consumables — the only recurring cost to the user is occasional battery recharging, which is effectively negligible.

Based on our prototype bill of materials, the estimated hardware cost is approximately LKR 6,550 (USD 20) per unit, excluding one-off development, tooling, and labour costs. With higher production volumes, component sourcing and manufacturing efficiencies are expected to reduce the per-unit hardware cost further.

At this price point, GlucoMonitor offers a significantly more affordable alternative to continuous glucose monitoring (CGM) systems while eliminating recurring sensor replacement costs. Although it is not intended to replace clinically approved CGMs for insulin dosing, GlucoMonitor is designed as an accessible solution for routine glucose trend monitoring, particularly for cost-sensitive users and individuals who lack access to the existing CGM technologies.

9. Sustainability

- No disposable consumables: unlike test strips, lancets, or replaceable CGM sensors, the optical clip generates no recurring medical waste.
- Rechargeable LiPo battery reduces single-use battery waste compared to disposable-battery wearables.
- Aggressive deep-sleep power management lowers overall energy consumption and extends battery (and therefore device) lifespan.
- The 3D-printed enclosure can be reprinted/repared rather than replaced if damaged, and the design is modular enough that individual components (sensor, battery, display) can be swapped without discarding the whole unit.

10. Ethical Considerations

- Non-diagnostic framing: because the device has not undergone clinical validation, all outputs at this stage must be clearly labelled as experimental/research-only to avoid patients making real treatment decisions (e.g. insulin dosing) based on reading.
- Data privacy: PPG and motion data are physiological and potentially sensitive; storage and any future transmission must follow the data-security practices.
- Research ethics: the literature this project builds on was collected from human participants under its own institutional ethical approval.
- Equity of access: the project's motivation of affordable and non-invasive monitoring is itself an ethical aim, intended to widen access to glucose monitoring for patients who currently cannot afford or tolerate existing options.

11. Conclusion and Future Work

GlucoMonitor demonstrates a feasible, low-cost architecture for non-invasive glucose estimation by combining a dual-wavelength PPG sensor, motion-artifact compensation, and an on-device TinyML model on commodity hardware. The circuit design, component selection, and TinyML integration strategy are grounded in current research literature.

Future work includes completing the PCB layout and enclosure fabrication, collecting bench-test data to validate signal quality and model accuracy, and — longer term — expanding into a companion mobile app, cloud-based trend tracking, and the clinical validation studies that would be required before any regulatory submission.

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